

SKILLS GUIDE



Innovation and creativity for designing parts and equipment that fulfill the client's brief.



Strong leadership skills to ensure that colleagues work to relevant electrical safety standards.



Good analytical skills to understand complex technical problems and devise cost-effective solutions.



High-level mathematical skills for recording, analyzing, and interpreting product test data.



Proficiency in using computer software and hardware when installing and fixing equipment.



TELECOMMUNICATIONS ENGINEER Specializes in the design and maintenance of electronic telecommunications technology, such as broadband, wireless networks, fiber-optic cabling, and satellite systems.



CONSULTING ENGINEER Provides advice to clients on the design and build of electrical systems and components, from power distribution to fire safety systems and interior lighting.



RESEARCH ENGINEER Works at a college or corporate research facility, carrying out research into emerging areas, such as nanoelectronics—electrical engineering on a molecular scale.

AT A GLANCE



YOUR INTERESTS Electrical circuitry • Engineering • Mathematics • Computing • Science • Physics • Optics • Technical drawing



ENTRY QUALIFICATIONS A degree in electrical engineering or a related subject is required. Internships or apprenticeships may be available.



LIFESTYLE Most electrical engineers work regular office hours, but evening, weekend, or on-call work may be required in some sectors.



LOCATION The work is based in an office or workshop, but visits to service equipment, monitor installations, or oversee manufacturing are common.



THE REALITIES Continual learning is required to keep pace with fast-changing technologies. Numerous career options are available.

RELATED CAREERS

- ▶ **SYSTEMS ANALYST** *see pp. 120–121*
- ▶ **NETWORK ENGINEER** *see pp. 124–125*
- ▶ **BROADCAST ENGINEER** Operates and maintains hardware and software used in television, radio, and other new-media broadcasts.
- ▶ **INFORMATION TECHNOLOGY (IT) CONSULTANT** Advises businesses on how to use IT systems to resolve operational issues.
- ▶ **ROBOTICS ENGINEER** Designs and builds robotic equipment for use in a variety of sectors, such as medical practice, drilling engineering, or motor-vehicle manufacturing.